

NATITINGOU, Benin

WMO: 653190

Lat: 10.317N

Lon: 1.383E

Elev: 461

StdP: 95.91

Time Zone: 1.00 (E01)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	16.4	17.8	-3.6	3.0	26.0	-1.9	3.4	26.3	6.1	26.5	5.4	26.7	0.3	140	0.288

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	11.8	37.8	22.4	36.9	22.9	36.0	22.9	28.5	33.7	27.7	32.9	27.1	32.3	2.5	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.2	24.3	32.4	26.4	23.2	31.5	25.7	22.2	30.7	95.7	33.8	91.9	33.1	88.7	32.4	34.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6.0	5.1	4.3	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	27.3	26.3	28.6	30.5	30.0	28.5	26.9	25.7	25.3	25.9	26.9	26.8	26.4
	DBStd	2.15	1.79	1.78	1.36	1.61	1.54	1.43	1.16	1.08	1.11	1.03	0.99	1.59
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6314	506	521	637	598	574	508	486	474	476	522	504	508
	CDD18.3	3272	247	287	378	348	316	258	227	216	226	264	254	250
	CDH23.3	35537	2797	3735	5342	4642	3817	2626	1713	1437	1628	2346	2711	2743
(14) Wind	WSAvg	1.5	1.6	1.5	1.6	1.9	1.8	1.6	1.6	1.4	1.0	1.1	1.2	1.4
	PrecAvg	1247	1	4	23	81	120	149	211	274	255	104	17	3
	PrecMax	1768	42	33	63	263	202	271	436	451	482	245	205	62
	PrecMin	830	0	0	0	17	37	20	72	79	111	16	0	0
	PrecStd	216	7	8	18	47	40	56	81	81	77	50	37	10
	DB	37.0	38.4	39.2	38.5	36.2	34.0	31.6	31.1	32.1	34.6	36.4	36.1	36.1
	MCWB	20.2	21.8	22.0	23.4	27.1	26.1	25.8	26.0	26.0	24.6	20.2	19.9	19.9
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	35.4	37.0	38.1	37.2	34.8	32.7	30.6	30.0	31.1	33.2	35.2	35.1	35.1
	2% MCWB	19.9	22.1	22.0	24.1	26.5	26.0	25.5	25.4	25.9	25.0	20.7	19.2	19.2
	5% DB	34.2	36.0	37.1	36.1	33.9	31.9	29.9	29.1	30.2	32.2	34.1	34.2	34.2
	5% MCWB	19.8	21.6	22.8	25.0	26.2	25.7	25.2	25.0	25.7	25.4	20.8	19.0	19.0
	10% DB	33.0	35.1	36.2	34.9	33.0	31.0	29.0	28.3	29.3	31.2	33.0	33.0	33.0
	10% MCWB	19.3	21.0	23.2	25.6	26.2	25.6	24.8	24.7	25.3	25.2	21.0	19.1	19.1
	WB	24.9	27.7	28.5	29.5	30.0	28.6	27.0	26.8	27.3	28.1	26.4	26.6	26.6
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	MCDB	32.8	35.6	35.7	35.1	34.6	32.0	30.2	30.1	30.7	32.2	31.8	32.7	32.7
	2% WB	23.5	25.9	27.5	28.3	28.5	27.3	26.2	26.0	26.6	27.1	25.5	24.7	24.7
	2% MCDB	32.5	34.4	35.1	33.9	33.0	31.2	29.6	29.2	30.0	31.5	31.0	31.4	31.4
	5% WB	22.2	24.1	26.7	27.4	27.5	26.6	25.6	25.4	26.1	26.4	24.7	22.9	22.9
	5% MCDB	31.9	32.8	34.2	33.2	32.2	30.6	29.0	28.5	29.5	30.8	30.3	30.5	30.5
	10% WB	20.8	23.0	25.8	26.6	26.7	25.9	25.1	24.9	25.5	25.7	23.8	21.3	21.3
	10% MCDB	31.1	32.2	33.3	32.6	31.6	29.9	28.4	27.8	28.8	30.0	29.3	29.7	29.7
(35) Mean Daily Temperature Range	MDBR	13.9	13.3	11.8	9.8	8.5	7.6	6.5	6.1	7.6	9.9	14.3	15.0	15.0
	5% DB	16.0	14.6	13.5	11.4	9.9	8.8	7.8	7.4	8.8	11.1	15.7	16.0	16.0
	5% WB	8.3	7.3	6.2	4.5	4.2	3.8	3.6	3.7	4.6	5.0	6.1	7.5	7.5
	MCDBR	15.4	12.9	10.9	9.6	8.9	8.2	7.5	7.1	8.4	10.2	12.8	15.1	15.1
	MCWBR	9.7	7.7	5.5	4.8	5.1	4.5	4.0	4.0	4.8	5.4	6.3	9.2	9.2
(40) Clear-Sky Solar Irradiance	taub	0.658	0.723	0.797	0.710	0.618	0.567	0.518	0.504	0.510	0.596	0.576	0.575	0.575
	taud	1.658	1.547	1.455	1.657	1.806	1.939	2.117	2.158	2.113	1.854	1.829	1.817	1.817
	Ebn at Noon	675	651	615	666	715	742	784	807	810	735	734	728	728
	Edh at Noon	249	287	319	258	218	188	158	154	163	208	208	208	208
(44) All-Sky Solar Radiation	RadAvg	5.40	5.61	5.84	5.91	5.73	5.30	4.72	4.45	4.95	5.60	5.63	5.38	5.38
	RadStd	0.20	0.20	0.23	0.18	0.20	0.16	0.17	0.22	0.21	0.21	0.14	0.20	0.20

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page